SLCRF: Subspace Learning With Conditional Random Field for Hyperspectral Image Classification

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***Abstract*— Subspace learning (SL) plays an essential role in hyperspectral image (HSI) classification since it can provide an effective solution to reduce the redundant information in the image pixels of HSIs. Previous works about SL aim to improve the accuracy of HSI recognition. Using a large number of labeled samples, related methods can train the parameters of the proposed solutions to obtain better representations of HSI pixels. However, the data instances may not be sufficient to learn a precise model for HSI classification in real applications. Moreover, it is well known that it takes much time, labor, and human expertise to label HSI images. To avoid the abovementioned problems, a novel SL method that includes the probability assumption called SL with the conditional random field (SLCRF) is developed. In SLCRF, the 3-D convolutional autoencoder (3DCAE) is first introduced to remove the redundant information in HSI pixels. Besides, the relationships are also constructed using spectral–spatial information among the adjacent pixels. Then, the conditional random field (CRF) framework can be constructed and further embedded into the HSI SL procedure with the semisupervised approach. Through the linearized alternating direction method termed LADMAP, the objective function of SLCRF is optimized using a defined iterative algorithm. The proposed method is comprehensively evaluated using the challenging public HSI data sets. We can achieve state-of-the-art performance using these HSI sets.**

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***Index Terms*— Conditional random field (CRF), hyperspectral image (HSI) classification, optimization, relationship construction, subspace learning (SL).**

I. Introduction

HYPERSPECTRAL image classification, as a hot and significant research topic, has received extensive attention in the fields of remote sensing and computer vision [1], [2]. The difficulty of estimating the relationship between two pixels is the prominent challenge in hyperspectral image (HSI) recognition due to its high dimensionality [3]–[8]. Hence, subspace learning (SL) plays an essential role in HSI classification since it is an effective solution to reduce the redundant information of HSI pixels [9].

The related works about SL have appeared in the fields of computer vision, as well as in remote sensing. In computer vision, there are a variety of SL models and approaches to transform the high-dimensional image data into more compact representations [10], [11]. Dimensionality reduction maps the original feature's high-dimensional data to a low-dimensional subspace, which can be equal to SL. Some classical approaches have been developed to effectively preserve the statistical properties, including principal component analysis (PCA) [12] and independent component analysis [13]. Manifold learning is usually adopted and embedded in the SL procedure [14]–[16]. To discover the nonlinear degrees of freedom that may underlie the complex natural observations, isometric feature mapping (ISOMAP) was developed in [14]. For the Laplacian eigenmap (LE), a geometrically motivated algorithm was proposed to provide a computationally efficient approach for SL [15]. Locally linear embedding was designed to find compact expressions of high-dimensional features using the unsupervised learning algorithm [16]. Moreover, a feature representation learning method was adopted by combining image understanding, feature correlation, and feature learning to learn the underlying subspace [10].

In the related field of computer vision, many SL approaches have been introduced to perform related tasks, including facial recognition, image understanding, and image segmentation. However, directly applying these methods to remote sensing may degrade the accuracy of remote sensing image recognition. Most importantly, the structures of the data sources are different. HSIs have distinctive data components and image features. Hence, a specially designed SL algorithm should

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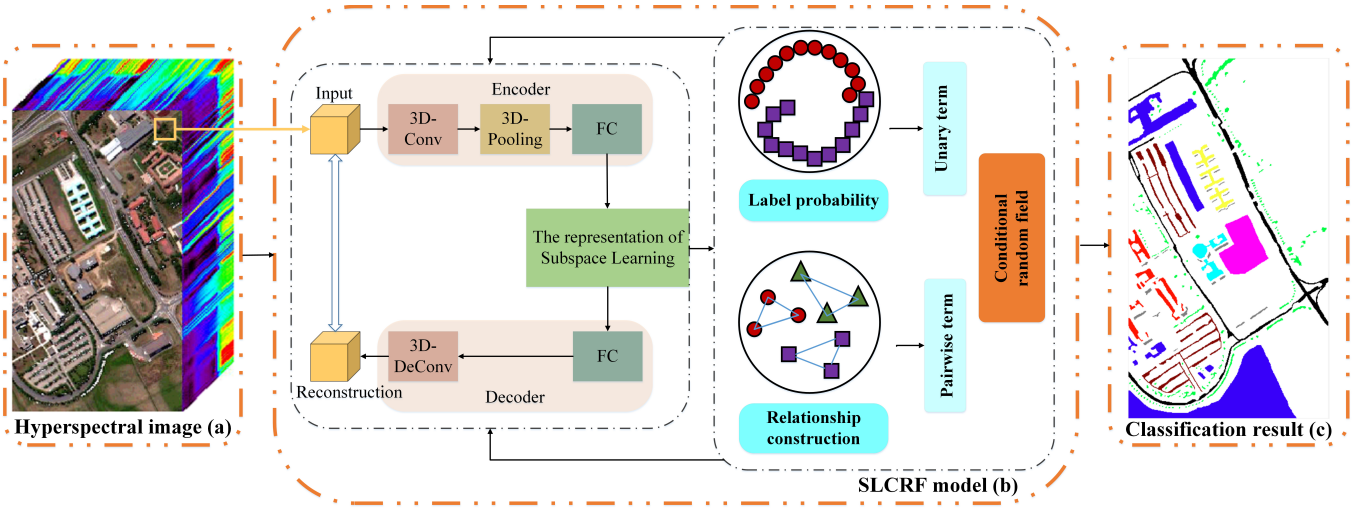


Fig. 1. SLCRF workflow for HSI classification. (a) HSI. (b) Learning procedure of the SLCRF. (c) Classification results. In (b), the SL refers to 3DCAE, which consists of an encoder and a decoder. The representations of the latent subspace with different classes are denoted as squares, triangles, and circles. Then, the CRF framework is constructed and further embedded into the SL procedure.

be constructed to better recognize HSIs. A dimensionality reduction method with an enhanced alignment technique was proposed to maximize the class distances and the features' inherent geometric structures using labeled and unlabeled training samples [9]. For the class separability, neighborhood data structures and nearby feature line measurements were introduced to determine the transformation of the dimensionality reduction in the eigenspace. To take advantage of the spectral information and the spatial correlation between HSI pixels and further avoid the "salt and pepper" problem, researchers developed the pixel-level and superpixel-level aware subspace learning (PSASL) method [17]. A similar method also appeared in work of [18] in which a projection learning approach with both local and global consistency constraints was proposed for aerial image recognition.

Previous works about SL aim at improving the accuracy of HSI recognition. With a lot of labeled training samples, some methods can train the parameters of the proposed solutions to obtain better representations of HSI pixels. Convolutional neural networks (CNNs) can be used for HSI classification by combining with the softmax classifier (SC) and cross-entropy loss in a supervised way. Hu *et al.* [19] employed deep CNNs using an SC for HSI classification in a pixel level. Lee and Kwon [20] proposed a wider and deeper CNN that has a multiscale convolutional filter to obtain initial spatial and spectral feature maps. The maps are then fed through a fully convolutional network that produces the labels of HSI with an SC. Such an end-to-end classification system can classify the HSI in a supervised way. However, supervised methods need large training samples. In real applications, the data instances may not be sufficient to learn a precise model for HSI recognition. Moreover, it is well known that it takes much time, labor, and human expertise to label HSI images [21].

Recent research on self-supervised learning [22]–[24] has shown us that data information can be considered as a self-supervision source for SL. In this way, better features of HSI pixels can be obtained without large amounts of manual labeling. The information on HSI data mainly includes the probabilities of pixels belonging to different categories and the relationships between the pixels. To better embed the above data information into our proposed method, a probability model can be introduced to obtain better representations of HSI pixels. CRF is a good way that can incorporate different image features for classification [25], [26]. In the work of [26], a saliency method was developed that jointly learns a CRF and a visual dictionary. The convolutional CRF using the confidence map was proposed to achieve better focus region detection [27]. The CRF advantage further extended by using CNNs for low-level feature extraction [28].

SL with the conditional random field (SLCRF) is developed to obtain the subspace of the HSI pixels using the semisupervised approach in this article. To increase the accuracy of the SL of HSIs, the CRF is embedded into the SLCRF learning framework. First, a 3-D convolutional autoencoder (3DCAE) is trained to remove the redundant information of HSI pixels through the reconstruction error. Thus, the latent subspace can be obtained from the original data. To obtain accurate subspace representation, the relationship matrix is further formulated by combining the relationships of the latent space and the spatial distance. However, good representation of the latent subspace and good relationships among pixels cannot be obtained without the probability maximization. The CRF framework can provide a good approach to enhance the performance of SL. The CRF unary and pairwise terms are constructed using the probabilities of pixels belonging to different categories and the relationships between the pixels. Then, a novel approach termed SLCRF is proposed to incorporate the CRF framework into the SL procedure. The SL, the relationship construction, and the semisupervised constraints included in the CRF can build a unified objective function. The objective function of SLCRF can further use an iterative optimization algorithm. The developed method is tested using three public adopted HSI data sets. The results can validate that the SLCRF can perform better than the related HSI classification approaches. The workflow of the SLCRF is shown in Fig. 1.

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TABLE I
NOTATIONS AND DEFINITIONS OF SLCRF

Notation	Definition
$\mathbf{X}$	Data matrix. $\mathbf{X} \in \mathbb{R}^{b\times b\times d}$
$\mathbf{X}^{(M)}$	The reconstructed matrix of $\mathbf{X}$. $\mathbf{X}^{(M)} \in \mathbb{R}^{b\times b\times d}$
$\boldsymbol{x}^{(M_1)}$	The representation in the latent subspace of $\mathbf{X}$.
$\mathbf{S}$	The matrix of the relationship between pixels. $\mathbf{S} \in \mathbb{R}^{n\times n}$
$\mathbf{Z}$	The matrix of the sparse representation. $\mathbf{Z} \in \mathbb{R}^{n\times n}$
θ	The parameters of CAE.
$\mathbf{Y}$	The predicted labels of pixels. $\mathbf{Y} \in \mathbb{R}^{c\times n}$
$\mathbf{h}_1$	The parameters of the unary term of the CRF.
$\mathbf{h}_2$	The scalar that measures the weight of the labeling smoothness.
$\mathcal{G}$	The adaptive graph.
$\alpha, \beta, \gamma, \eta$	Balance the corresponding terms.
λ_1, λ_2	Balance the corresponding terms.
d	The number of bands.
n	The number of samples.
P_i	The position of pixel i.

The main contributions of the SLCRF can be further summarized in the following.

1) With the learned subspace, the probabilities of pixels belonging to different categories and the relationships between the neighboring pixels are constructed to form the corresponding terms of the CRF. With the CRF, the better HSI subspace can be obtained to alleviate the need for labeled samples.
2) SLCRF integrates the SL, the relationship construction, and the semisupervised constraints that are defined in the CRF to form the objective function. An iterative optimization algorithm is used to solve the objective function of the SLCRF.
3) SLCRF is an efficient end-to-end trained framework. 3DCAE allows for SL. The CRF is further embedded into SL to provide feedback information for better SL. It can outperform state-of-the-art approaches using three image data sets.

II. SUBSPACE LEARNING WITH THE CONDITIONAL RANDOM FIELD

In this section, we first present the motivation of this article in Section II-A. Second, we introduce the definition of SL in Section II-B. Next, the relationships' construction with SL is described in Section II-C and the CRF introduction for SL is described in Section II-D. Moreover, the objective function of SLCRF is formulated in Section II-E.

A. Motivation

With certain amount of labeled training samples, the related SL methods can obtain better representations of HSI pixels. However, the data instances may not be sufficient in real applications. Semisupervised methods can alleviate the need for a large number of labeled samples. In this article, we propose the SLCRF method to obtain the subspace of the HSI pixels in a semisupervised way. The representation of the latent subspace is obtained by training the 3DCAE. To obtain accurate subspace representation, the relationship matrix is further formulated by combining the relationships of the latent space and the spatial distance. However, good representation of the latent subspace and good relationships among pixels cannot be obtained without the probability maximization. The CRF can provide a better approach to enhance the SL's performance, which is incorporated into the SL procedure. SLCRF integrates SL, the relationship construction, and the semisupervised constraints included in the CRF into a unified objective function. An iterative optimization algorithm is used to solve the objective function of the SLCRF.

B. Definition of Subspace Learning

Given an HSI, the high-dimensional HSI data set can be constructed in a mathematical format. Let $P_{(x,y)} \in \mathbb{R}^{d\times 1}$ be a pixel located at (x, y) of the HSI, where d is the number of spectrum bands. The spatial neighborhood of $P_{(x,y)}$ is defined as $\mathbf{X} \in \mathbb{R}^{b\times b\times d}$, where $b \times b$ is the spatial neighborhood of pixel (x, y). We use the 3DCAE framework to obtain its representation in the latent subspace, which consists of an encoder and a decoder. The encoder maps the original data $\mathbf{X} \in \mathbb{R}^{b\times b\times d}$ to the representation of latent subspace $\boldsymbol{x}^{(M_1)} \in \mathbb{R}^{K\times 1}$, where K represents the dimensions of the latent subspace. The decoder reconstructs an approximation $\mathbf{X}^{(M)} \in \mathbb{R}^{b\times b\times d}$ of $\mathbf{X}$ from $\boldsymbol{x}^{(M_1)}$. There have M layers in the 3DCAE, where the encoder has M_1 layers and decoder has $(M - M_1)$ layers. The encoder consists of 3-D convolutional layers, 3-D max-pooling layers, and fully connected layers to learn the representations of latent subspace. The decoder consists of 3-D deconvolution layers and fully connected layers to reconstruct the original data. More details of 3DCAE can be referred to Appendix A.

The objective function of 3DCAE can be formulated as follows:

$$\Theta_1(\theta) = \frac{1}{b \times b \times d} \sum_{x=0}^{b-1} \sum_{y=0}^{b-1} \sum_{z=0}^{d-1} \left(X_{x,y,z} - X_{x,y,z}^{(M)}\right)^2 + \frac{\alpha}{2} \|\mathbf{W}\|_F^2 \tag{1}$$

where α is the balancing term between the reconstruction error and the weight decay term. θ represents the set of all parameters in 3DCAE, and $\mathbf{W}$ consists of the weights in all layers. $X_{x,y,z}$ represents the value at position (x, y, z) of the input $\mathbf{X}$, while $X_{x,y,z}^{(M)}$ represents the reconstructed value at position (x, y, z). The weight decay term helps to decrease the magnitude of the weights and prevent overfitting. Based on the 3DCAE [29], [30], (1) acquires the latent subspace from the original data $\mathbf{X}$ by training the 3DCAE through the reconstruction error. By minimizing the equation, we can acquire the subspace of the corresponding HSI.

C. Relationships' Construction With Subspace Learning

To obtain accurate subspace representation, the relationship matrix $\mathbf{S}$ between the pixels is constructed to enhance the SL, which is formulated by combining the relationships of the latent space and the spatial distance. The relationships of

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the latent space are constructed by using the sparse coding to obtain the sparse matrix, which describes the relationships in the latent subspace. Let $\mathbf{X}^{(M_1)} = [\boldsymbol{x}_1^{(M_1)}, \boldsymbol{x}_2^{(M_1)}, \ldots, \boldsymbol{x}_n^{(M_1)}] \in \mathbb{R}^{K\times n}$ be the representation of latent subspace for all HSI pixels, where n is the number of pixels. Considering $\mathbf{X}^{(M_1)}$ as the dictionary, the following equation is utilized to implement the sparse coding of latent subspace:

$$\Theta_2(\theta, \mathbf{Z}) = \left\|\mathbf{X}^{(M_1)} - \mathbf{X}^{(M_1)}\mathbf{Z}\right\|_F^2 + \beta\|\mathbf{Z}\|_1 \tag{2}$$

where β is the balancing constant, $\|\cdot\|_F^2$ represents the Frobenius norm of a matrix, $\|\cdot\|_1$ represents the $L1$ norm of a matrix. $\mathbf{Z}$ describes the relationships the latent subspace, and each column vector of $\mathbf{Z}$ is the sparse representation. To better estimate $\mathbf{Z}$, the initialization of $\mathbf{Z}$ is obtained by optimizing the equation $\|\mathbf{X}^{(M_1-1)} - \mathbf{X}^{(M_1-1)}\mathbf{Z}\|_F^2 + \beta\|\mathbf{Z}\|_1$ with $\mathbf{X}^{(M_1-1)}$ fixed. The relationship matrix $\mathbf{S}_1$ of the latent space can be formed as follows:

$$\mathbf{S}_1 = |\mathbf{Z} + \mathbf{Z}^T|/2. \tag{3}$$

Second, the relationship matrix $\mathbf{S}_2$ of spatial distance is constructed to describe the relationships of the spatial distance

$$\mathbf{S}_2 = \begin{cases} \exp\left(-\dfrac{\left\|P_i - P_j\right\|_2^2}{\omega}\right), & \boldsymbol{x}_i^{(M_1)} \in N\left(\boldsymbol{x}_j^{(M_1)}\right) \text{ or} \\ & \boldsymbol{x}_j^{(M_1)} \in N\left(\boldsymbol{x}_i^{(M_1)}\right) \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where $N(\boldsymbol{x}_j^{(M_1)})$ is the k-nearest neighbors of $\boldsymbol{x}_i^{(M_1)}$ that are constructed using the spatial distance $\left\|P_i - P_j\right\|_2$. ω is an alternative parameter.

By combining the relationships of the latent space and the spatial distance, we can formulate the relationship matrix $\mathbf{S}$ to denote the relationships in subspace

$$\mathbf{S} = \mathbf{S}_1 + \gamma\,\mathbf{S}_2 \tag{5}$$

where γ is a constant.

D. CRF Introduction for Subspace Learning

From the definition of the SL, we can obtain the representation in the latent subspace. However, good representation of the latent subspace and good relationships among pixels cannot be obtained without the probability maximization. The CRF can provide a better approach to enhance the performance of SL.

The CRF usually needs a graph to construct its terms, including the unary term and binary term. In our method, the graph $\mathbf{G} = (\nu, \varepsilon)$ is built to reflect the relationships among different HSI pixels. ν refers to the representations of the latent subspace in an HSI. Each vertex corresponds to one representation of the latent subspace. ε is the edges that are linked by the neighboring pixels. Its weights can be constructed using the matrix $\mathbf{S}$ defined in (5).

In a CRF, the unary term describes the possibility of one belonging to one certain category, and the pairwise term refers to the relationships between different pixels. The CRF $(\mathbf{X}^{(M_1)}, \mathbf{Y})$ can be formulated by a Gibbs distribution

$$p(\mathbf{Y}|\mathbf{X}^{(M_1)}; \theta, \mathbf{Z}, \mathbf{h}) = \frac{1}{C}\exp\left(-\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu} E(\boldsymbol{x}_i^{(M_1)}, \theta, \mathbf{Z}, \mathbf{h})\right) \tag{6}$$

where C is the partition function and $E(\theta, \mathbf{Z}, \mathbf{h})$ is the energy function with the weight $\mathbf{h}$, parameters θ, and $\mathbf{Z}$. To optimize the model, it needs to maximize conditional distribution $p(\mathbf{Y}|\mathbf{X}^{(M_1)}; \theta, \mathbf{Z}, \mathbf{h})$

$$\max_{\theta,\mathbf{Z},\mathbf{h}} p(\mathbf{Y}|\mathbf{X}^{(M_1)}; \theta, \mathbf{Z}, \mathbf{h}) = \max_{\theta,\mathbf{Z},\mathbf{h}} \frac{1}{C}\exp\left(-\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu} E(\boldsymbol{x}_i^{(M_1)}, \theta, \mathbf{Z}, \mathbf{h})\right) \tag{7}$$

which is equivalent to

$$\min_{\theta,\mathbf{Z},\mathbf{h}} \sum_{\boldsymbol{x}_i^{(M_1)}\in\nu} E(\boldsymbol{x}_i^{(M_1)}, \theta, \mathbf{Z}, \mathbf{h}). \tag{8}$$

Then, we can formulate a novel CRF model

$$E(\boldsymbol{x}_i^{(M_1)}, \theta, \mathbf{Z}, \mathbf{h}) = \phi(\theta, \mathbf{h}_1) + \eta\varphi(\mathbf{Z}, \mathbf{h}_2) \tag{9}$$

where η is the balancing constant, $\mathbf{h} = \{\mathbf{h}_1, \mathbf{h}_1\}$ is the weight matrix, $\mathbf{h}_1$ is adopted to compute the categorical possibilities using the softmax function, and $\mathbf{h}_2$ measures the weight of the pairwise term. The function ϕ is the unary term in the CRF

$$\phi(\theta, \mathbf{h}_1) = \sum_{\boldsymbol{x}_i^{(M_1)}\in\nu} -I(t)\log\left(p\left(\boldsymbol{y}_i|\boldsymbol{x}_i^{(M_1)}\right)\right) \tag{10}$$

where $\mathbf{h}_1 = \{\mathbf{W}^{\mathbf{h}_1}, \mathbf{b}^{\mathbf{h}_1}\}$. The value of $I(t)$ is 1 if t equals to the desired label of pixel i; otherwise, its value is 0. $p(\boldsymbol{y}_i|\boldsymbol{x}_i^{(M_1)})$ is achieved by using the softmax function Φ

$$p\left(\boldsymbol{y}_i|\boldsymbol{x}_i^{(M_1)}\right) = \Phi\left(\mathbf{W}^{\mathbf{h}_1}\boldsymbol{x}_i^{(M_1)} + \mathbf{b}^{\mathbf{h}_1}\right). \tag{11}$$

The functions φ is the pairwise term in the CRF

$$\varphi(\mathbf{Z}, \mathbf{h}_2) = \sum_{\boldsymbol{x}_i^{(M_1)}\in\nu} \sum_{\boldsymbol{x}_j^{(M_1)}\in N(\boldsymbol{x}_i^{(M_1)})} -\mathbf{h}_2\mathbf{S}_{i,j} \tag{12}$$

where $\mathbf{h}_2$ is scalar measuring the weight of labeling smoothness

$$\mathbf{h}_2 = \frac{(\boldsymbol{y}_i - \boldsymbol{y}_j)^2}{\sum_{\boldsymbol{x}_j^{(M_1)}\in N(\boldsymbol{x}_i^{(M_1)})} \mathbf{S}_{i,j}}. \tag{13}$$

E. Objective Function Formulation of SLCRF

Given the training samples $\mathbf{X}$ and the corresponding predicated labels, the SLCRF method is to learn the SL parameters θ and the weights $\mathbf{h}$ in the CRF to maximize the joint likelihood of training samples. Here, we combine all the terms together and establish the SLCRF model

$$\min_{\theta,\mathbf{Z},\mathbf{h}} \Theta_1(\theta) + \lambda_1\Theta_2(\theta, \mathbf{Z}) + \lambda_2 E(\theta, \mathbf{Z}, \mathbf{h}) \tag{14}$$

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where $\lambda_1 > 0$ and $\lambda_2 > 0$ are the tradeoff parameters. Equivalently, the SLCRF model can be defined using the following equation:

$$\begin{aligned}\min_{\theta,\mathbf{Z},\mathbf{h}} &\frac{1}{b\times b\times d}\sum_{x=0}^{b-1}\sum_{y=0}^{b-1}\sum_{z=0}^{d-1}\left(X_{x,y,z}-X^{(M)}_{x,y,z}\right)^2+\frac{\alpha}{2}\|\mathbf{W}\|_F^2\\&+\lambda_1\left\|\mathbf{X}^{(M_1)}-\mathbf{X}^{(M_1)}\mathbf{Z}\right\|_F^2+\lambda_1\beta\|\mathbf{Z}\|_1\\&+\lambda_2\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu}-I(t)\log\left(p(\boldsymbol{y}_i|\boldsymbol{x}_i^{(M_1)})\right)\\&+\lambda_2\eta\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu}\sum_{\boldsymbol{x}_j^{(M_1)}\in N(\boldsymbol{x}_i^{(M_1)})}-\mathbf{h}_2\mathbf{S}_{i,j}.\end{aligned}\tag{15}$$

Due to the close dependence of $\mathbf{S}$ on $\mathbf{Z}$, the auxiliary matrix $\mathbf{M}\in\mathbb{R}^{n\times n}$ is introduced to separate (15) Then, the whole objective function can be transferred into

$$\begin{aligned}\min_{\theta,\mathbf{Z},\mathbf{h}} &\frac{1}{b\times b\times d}\sum_{x=0}^{b-1}\sum_{y=0}^{b-1}\sum_{z=0}^{d-1}\left(X_{x,y,z}-X^{(M)}_{x,y,z}\right)^2+\frac{\alpha}{2}\|\mathbf{W}\|_F^2\\&+\lambda_1\left\|\mathbf{X}^{(M_1)}-\mathbf{X}^{(M_1)}\mathbf{M}\right\|_F^2+\lambda_1\beta\|\mathbf{M}\|_1\\&+\lambda_2\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu}-I(t)\log\left(p\left(\boldsymbol{y}_i|\boldsymbol{x}_i^{(M_1)}\right)\right)\\&+\lambda_2\eta\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu}\sum_{\boldsymbol{x}_j^{(M_1)}\in N(\boldsymbol{x}_i^{(M_1)})}-\mathbf{h}_2\mathbf{S}_{i,j}\\&\text{s.t. }\mathbf{Z}=\mathbf{M}.\end{aligned}\tag{16}$$

III. Optimization of SLCRF

To optimize the SLCRF model, the linearized alternating direction method termed LADMAP [31] is adopted. The corresponding augmented Lagrangian function of the optimization problem defined in (16) can be written in the following equation:

$$\begin{aligned}&L(\theta,\mathbf{Z},\mathbf{M},\mathbf{h},\mathbf{T})\\&=\frac{1}{b\times b\times d}\sum_{x=0}^{b-1}\sum_{y=0}^{b-1}\sum_{z=0}^{d-1}\left(X_{x,y,z}-X^{(M)}_{x,y,z}\right)^2\\&+\frac{\alpha}{2}\|\mathbf{W}\|_F^2+\lambda_1\left\|\mathbf{X}^{(M_1)}-\mathbf{X}^{(M_1)}\mathbf{M}\right\|_F^2+\lambda_1\beta\|\mathbf{M}\|_1\\&+\lambda_2\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu}-I(t)\log\left(p\left(\boldsymbol{y}_i|\boldsymbol{x}_i^{(M_1)}\right)\right)\\&+\lambda_2\eta\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu}\sum_{\boldsymbol{x}_j^{(M_1)}\in N(\boldsymbol{x}_i^{(M_1)})}-\mathbf{h}_2\mathbf{S}_{i,j}\\&+\langle\mathbf{T},\mathbf{Z}-\mathbf{M}\rangle+\frac{\epsilon}{2}\|\mathbf{Z}-\mathbf{M}\|_F^2\end{aligned}\tag{17}$$

which can be transferred into

$$\begin{aligned}&L(\theta,\mathbf{Z},\mathbf{M},\mathbf{h},\mathbf{T})\\&=\frac{1}{b\times b\times d}\sum_{x=0}^{b-1}\sum_{y=0}^{b-1}\sum_{z=0}^{d-1}\left(X_{x,y,z}-X^{(M)}_{x,y,z}\right)^2\\&+\frac{\alpha}{2}\|\mathbf{W}\|_F^2+\lambda_1\left\|\mathbf{X}^{(M_1)}-\mathbf{X}^{(M_1)}\mathbf{M}\right\|_F^2+\lambda_1\beta\|\mathbf{M}\|_1\\&+\lambda_2\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu}-I(t)\log\left(p\left(\boldsymbol{y}_i|\boldsymbol{x}_i^{(M_1)}\right)\right)\\&+\lambda_2\eta\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu}\sum_{\boldsymbol{x}_j^{(M_1)}\in N(\boldsymbol{x}_i^{(M_1)})}-\mathbf{h}_2\mathbf{S}_{i,j}\\&+\frac{\epsilon}{2}\left(\left\|\mathbf{Z}-\mathbf{M}+\frac{\mathbf{T}}{\epsilon}\right\|_F^2\right)-\frac{1}{2\epsilon}\left(\|\mathbf{T}\|_F^2\right)\end{aligned}\tag{18}$$

where $\mathbf{T}$ is the Lagrangian multiplier and $\epsilon > 0$ is a penalty parameter.

θ can be optimized when $\mathbf{Z}$, $\mathbf{M}$, and $\mathbf{h}$ are fixed. The problem that is defined in (18) can be further rewritten as (19). For θ, we can optimize the following problem:

$$\begin{aligned}\min_{\theta}L(\theta)=\min_{\theta}&\frac{1}{b\times b\times d}\sum_{x=0}^{b-1}\sum_{y=0}^{b-1}\sum_{z=0}^{d-1}\left(X_{x,y,z}-X^{(M)}_{x,y,z}\right)^2\\&+\frac{\alpha}{2}\|\mathbf{W}\|_F^2+\lambda_1\left\|\mathbf{X}^{(M_1)}-\mathbf{X}^{(M_1)}\mathbf{Z}\right\|_F^2+\lambda_1\beta\|\mathbf{Z}\|_1.\end{aligned}\tag{19}$$

$\mathbf{Z}$ is computed when θ, $\mathbf{M}$, and $\mathbf{h}$ are fixed. As for $\mathbf{Z}$, we can optimize the following problem:

$$\begin{aligned}\min_{\mathbf{Z}}L(\mathbf{Z})=\min_{\mathbf{Z}}&\lambda_1\left\|\mathbf{X}^{(M_1)}-\mathbf{X}^{(M_1)}\mathbf{Z}\right\|_F^2\\&+\lambda_2\eta\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu}\sum_{\boldsymbol{x}_j^{(M_1)}\in N(\boldsymbol{x}_i^{(M_1)})}-\mathbf{h}_2\mathbf{S}_{i,j}+\frac{\epsilon}{2}\left\|\mathbf{Z}-\mathbf{M}+\frac{\mathbf{T}}{\epsilon}\right\|_F^2.\end{aligned}\tag{20}$$

$\mathbf{M}$ is computed when θ, $\mathbf{Z}$, and $\mathbf{h}$ are fixed. As for $\mathbf{M}$ subproblem, we can optimize the following problem:

$$\min_{\mathbf{M}}L(\mathbf{M})=\min_{\mathbf{M}}\beta\|\mathbf{M}\|_1+\frac{\epsilon}{2}\left\|\mathbf{Z}-\mathbf{M}+\frac{\mathbf{T}}{\epsilon}\right\|_F^2.\tag{21}$$

The Lagrangian multiplier $\mathbf{T}$ can be updated by

$$\mathbf{T}_{new}=\mathbf{T}_{old}+\epsilon(\mathbf{Z}-\mathbf{M}).\tag{22}$$

For the $\mathbf{h}$ optimization, with the original objective function that is defined in (18), we can get the following equations:

$$\begin{aligned}\min_{\mathbf{h}}L(\mathbf{h})=\min_{\mathbf{h}}&\lambda_2\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu}-I(t)\log\left(p(\boldsymbol{y}_i|\boldsymbol{x}_i^{(M_1)})\right)\\&+\lambda_2\eta\sum_{\boldsymbol{x}_i^{(M_1)}\in\nu}\sum_{\boldsymbol{x}_j^{(M_1)}\in N(\boldsymbol{x}_i^{(M_1)})}-\mathbf{h}_2\mathbf{S}_{i,j}.\end{aligned}\tag{23}$$

More details of SLCRF optimization can be referred to Appendix B. In summary, we can form Algorithm 1 as follows.

IV. Experiments

In this section, the performance of the proposed SLCRF approach for HSI classification is evaluated using different data sets. First, a brief description of the three used HSI data sets is presented, and then, the classification results of the SLCRF with some related methods are further compared.

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Algorithm 1 SLCRF

Input:

Input: Training data $\mathbf{X}$, parameters: β, γ, η, λ_1, λ_2. Initialize: θ, $\mathbf{Z}$, $\mathbf{h}$, $\mathbf{M}$, and $\mathbf{T}$; $\omega = 10^3$, $\epsilon = 0.01$, $\alpha = 0.0005$, $\delta_1 = 0.001$, $\delta_2 = 1$, $\tau = 0.0002$.

Output:

The optimal solutions θ, $\mathbf{Z}$, $\mathbf{h}$, $\mathbf{M}$, and $\mathbf{T}$.

While stopping criterion cannot meet **do**
Update θ using Eq. (19);
Update $\mathbf{Z}$ using Eq. (20);
Update auxiliary matrix $\mathbf{M}$ using Eq. (21);
Update $\mathbf{h}$ using Eq. (23);
Update the Lagrangian multiplier $\mathbf{M}$ using Eq. (22);

TABLE II
INDIAN PINES DATA SET

Class	Land Cover Type	Training (10%)	Labeling (5%)	Total
1	Alfalfa	5	2	46
2	Corn-notill	143	71	1,428
3	Corn-mintill	83	42	830
4	Corn	24	12	237
5	Grass-pasture	48	24	483
6	Grass-trees	73	36	730
7	Grass-pasture-mowed	3	1	28
8	Hay-windrowed	48	24	478
9	Oats	2	1	20
10	Soybean-notill	97	49	972
11	Soybean-mintill	245	123	2,455
12	Soybean-clean	59	29	593
13	Wheat	20	10	205
14	Woods	126	63	1,265
15	Bldg-grass-trees	39	20	386
16	Stone-Steel-Towers	9	5	93
	Total	1,024	512	10,249

A. Experimental Data Sets

1) Indian Pines Data Set: The AVIRIS sensor collected the corresponding data at the Indian Pines site in 1992. The image size is approximately 145 × 145. The data set consists of 224 spectral bands, and the spatial resolution is 20 m. The wavelength range is 0.4–2.5 μm. The bands of the water absorption region are discarded and 200 out of the 224 bands are kept in the experiment. The details of the Indian Pines data set that covers 16 classes and 10 249 pixels are shown in Table II.

2) University of Pavia Data Set (PaviaU): These data from Northern Italy was acquired for an urban area using the ROSIS-03 sensor. The HSI size is approximately 610 × 340, whereas its spatial resolution is approximately 1.3 m. In the experiment, 103 out of the 115 bands are used, and the water absorption bands are discarded. As a result, the details of the data set that covers 9 classes and 42 776 labeled samples are shown in Table III.

3) Houston Data Set (2018): The Houston data were organized by the Image Analysis and Data Fusion Technical Committee. The size of HSI is approximately 601 × 2,384 pixels and 48 spectral bands, whereas the spatial resolution is approximately 1.0 m. The details of the Houston data set that includes 20 categories and 504 712 labeled pixels are shown in Table IV.

TABLE III
PAVIAU DATA SET

Class	Land Cover Type	Training (5%)	Labeling (1%)	Total
1	Asphalt	345	66	6,631
2	Meadows	917	186	18,649
3	Gravel	131	20	2,099
4	Trees	156	30	3,064
5	Metal sheets	77	13	1,345
6	Bare Soil	246	50	5,029
7	Bitumen	60	13	1,330
8	Bricks	181	36	3,682
9	Shadows	44	94	947
	Total	2,157	508	42,776

TABLE IV
HOUSTON DATA SET

Class	Land Cover Type	Training (5%)	Labeling (1%)	Total
1	Healthy Grass	490	98	9,799
2	Stressed Grass	1,625	325	32,502
3	Artificial Turf	34	7	684
4	Evergreen Trees	679	136	13,588
5	Deciduous Trees	252	50	5,048
6	Bare Earth	225	45	4,516
7	Water	13	2	266
8	Residential Buildings	1,988	397	39,762
9	Non-residential Buildings	11,184	2,237	223,684
10	Roads	2,290	458	45,810
11	Sidewalks	1,700	340	34,002
12	Crosswalks	75	15	1,516
13	Major Thoroughfares	2,318	463	46,358
14	Highways	492	98	9,849
15	Railways	347	69	6,937
16	Paved Parking Lots	574	115	11,475
17	Unpaved Parking Lots	7	1	149
18	Cars	333	65	6,578
19	Trains	267	53	5,365
20	Stadium Seats	346	68	6,824
	Total	25,239	5,042	504,712

The 5% training samples for the PaviaU and Houston data sets and the 10% training samples for the Indian Pines data set are shown in Tables II–IV. To evaluate our proposed method compared to the related methods, the HSIs are classified under different percentages of training pixels, including 5, 10, 15, and 20.

B. Related Approaches

The developed method SLCRF is semisupervised for the relationship construction between the pixels in HSIs. The following approaches are adopted to compare with the SLCRF to the accuracy of the HSI classification.

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1) *PCA [12] and SC (PCA-SC):* First, the PCA is adopted to reduce the dimensionality of the spectral dimension of HSI. The neighborhood region of the pixel in the condensed data is extracted and flattened into a vector. The SC is used to classify the flattened vector with the labeled data.
2) *Locality Preserving Projections [32] and SC (LPP-SC):* LPP is used to map the HSI into a low dimension. The neighborhood region of the pixel in the low dimension is fed into a two-layer CNN. The SC is used to classify the spatial–spectral features with the labeled data.
3) *Convolutional autOencoder [33] and SC (CAE-SC):* CAE is a network that can be used to learn features of HSI in an unsupervised way. The CAE consists of one encoder function and one decoder function, and the encoder and decoder each have one convolutional layer. The encoder function maps the spatial–spectral input to a new feature representation of input data. Then, SC is used to classify the features extracted by CAE with the labeled data.
4) *Stacked CAE [30] and SC (SCAE-SC):* The SCAE is a network that consists of two shallow CAEs. The outputs of the hidden layer of CAE are wired to the input of subsequent CAE. The SC classifies the spatial–spectral features with the labeled data.
5) *3-D Convolutional Autoencoder [29] and CRF (3DCAE-CRF):* The operations of 3DCAE are all 3-D convolution, 3-D pooling, and 3-D batch normalization to explore spatial–spectral information. The features learned by 3DCAE are flattened into a 1-D vector. The CRF is used to produce labels of the 1-D vector with the labeled data, which is implemented by using the GraphCRF and NSlackSSVM tools of the PyStruct.

Moreover, to obtain the best HSI classifications, the following settings need to be further assessed. For the three data sets, s training samples are randomly selected as the labeled data. For the Indian Pines data set, the labeled data are set 5%, 6%, 7%, 8%, 9%, and 10% for each class, and for the PaviaU and Houston data set, the labeled data are set 1%, 2%, 3%, 4% and 5% for each class. The remaining HSI data are adopted for the classification.

In our SLCRF method, it selects k from the set {3, 4, 5, 6, 7, 8, 9, 10} to construct the spatial graph. The balancing parameters, such as λ_1 and λ_2, are tuned using the set $\{10^{-9}, 10^{-8}, \ldots, 10^{8}, 10^{9}\}$.

In the CAE, SCAE, 3DCAE, and our SLCRF methods, the number of feature dimensions is selected from the set $\{d/4, d/3, d/2, d, d \times 2, d \times 3, d \times 4\}$, where d is the number of spectrum bands. The number of the batch size is selected from the set {16, 32, 64, 128, 256}. The learning rate is selected from the set {0.01, 0.001, 0.0001, 0.0002, 0.00001, 0.00005}. The main architectures of SLCRF for the three data sets are shown in Tables V–VII.

TABLE V
ARCHITECTURES OF SLCRF FOR THE INDIAN PINES DATA SET

Encoder	Conv1	24×3×3×24	1×1×1	Yes	ReLU
	Conv2	24×3×3×48	20×1×1	Yes	ReLU
	FC1	432×216	1	No	ReLU
	FC2	216×144	1	No	ReLU
Decoder	FC3	144×216	1	No	ReLU
	FC4	216×432	1	No	ReLU
	Deconv1	9×3×3×24	22	Yes	ReLU
	Deconv2	27×3×3×24	1	Yes	ReLU

BN denotes the batch normalization, and FC denotes fully connection.

TABLE VI
ARCHITECTURES OF SLCRF FOR THE PAVIAU DATA SET

	Layer	Convolution	Srides	BN	Activation
Encoder	Conv1	11×3×3×24	1	Yes	ReLU
	Conv2	11×3×3×48	1	Yes	ReLU
	Max.Pool	9×1×1	9	No	-
	FC1	432×216	1	No	ReLU
Decoder	FC2	216×432	1	No	ReLU
	Deconv1	9×3×3×24	10	Yes	ReLU
	Deconv2	9×3×3×24	1	Yes	ReLU

BN denotes the batch normalization, and FC denotes fully connection.

TABLE VII
ARCHITECTURES OF SLCRF FOR THE HOUSTON DATA SET

	Layer	Convolution	Srides	BN	Activation
Encoder	Conv1	5×3×3×24	1	Yes	ReLU
	Conv2	5×3×3×48	1	Yes	ReLU
	Max.Pool	4×1×1	9	No	-
	FC1	480×240	1	No	ReLU
Decoder	FC2	240×480	1	No	ReLU
	Deconv1	4×3×3×24	4	Yes	ReLU
	Deconv2	9×3×3×24	1	Yes	ReLU

BN denotes the batch normalization, and FC denotes fully connection.

C. Evaluation Results

The HSI classification performances of the different methods are computed and compared in Tables VIII–XI and Figs. 2–5 using three metrics, such as the overall accuracy (OA), the average accuracy (AA), as well as the Kappa coefficient.

We can further obtain the following observations from Tables VIII–XI and Figs. 2–5.

1) Compared with the related methods, including the PCA-SC, LPP-SC, CAE-SC, SCAE-SC, and 3DCAE-CRF, the SLCRF can obtain the best HSI classification results for the testing image samples using the three data sets, which are much better than those that are acquired by other methods. The results further show that the SLCRF is very appropriate for image classification.
2) From Figs. 2–4, the HSI classification results achieved by the SLCRF using the three data sets are more compact than the related methods. The main reason is that SLCRF can well build the relationships between image pixels. These good relationships are very beneficial for image classification.
3) In Fig. 5, as the number of the labeled image pixels increases, the image classification accuracies of all approaches are improved. However, the SLCRF can obtain more accurate HSI overall classifications using different numbers of labeled image samples, further

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TABLE VIII
SEMISUPERVISED CLASSIFICATION RESULTS (%) WITH 5% LABELED SAMPLES FOR THE INDIAN PINES DATA SET

Class	PCA-SC	LPP-SC	CAE-SC	SCAE-SC	3DCAE-CRF	SLCRF
Alfalfa	9.30	20.93	8.89	0.00	49.05	85.71
Corn-notill	66.52	49.04	66.54	73.59	75.18	90.03
Corn-mintill	50.38	51.90	57.63	53.18	70.51	87.33
Corn	30.67	10.22	26.32	32.44	65.00	81.65
Grass-pasture	78.17	87.55	75.65	88.08	64.67	92.22
Grass-trees	95.82	86.15	96.22	95.09	89.53	95.16
Grass-pasture-mowed	11.54	19.23	0.00	11.11	27.54	54.55
Hay-windrowed	100.00	96.92	98.66	99.56	95.71	98.85
Oats	21.05	15.79	31.58	72.22	29.27	72.22
Soybean-notill	57.20	53.52	64.60	68.39	79.91	86.93
Soybean-mintill	79.50	80.92	77.46	79.93	85.45	91.42
Soybean-clean	22.20	29.66	41.52	38.74	75.38	83.62
Wheat	100.00	87.11	97.47	86.32	72.02	96.81
Woods	95.17	91.01	94.95	94.52	92.22	96.41
Bldg-grass-trees	54.64	55.74	40.55	48.79	63.12	81.63
Stone-Steel-Towers	78.41	86.36	82.95	82.23	90.00	95.40
OA	71.62	68.27	72.66	74.80	79.62	**90.70**
AA	59.41	55.63	60.06	63.89	70.28	**86.87**
Kappa	0.671	0.634	0.686	0.710	0.768	**0.894**

The best results are highlighted in bold.

TABLE IX
SEMISUPERVISED CLASSIFICATION RESULTS (%) WITH 1% LABELED SAMPLES FOR THE PAVIAU DATA SET

Class	PCA-SC	LPP-SC	CAE-SC	SCAE-SC	3DCAE-CRF	SLCRF
Asphalt	90.45	81.67	94.09	89.49	81.79	95.29
Meadows	96.42	93.81	96.98	96.95	91.78	97.83
Gravel	79.59	62.25	78.25	78.30	83.20	86.52
Trees	83.25	78.04	88.43	81.59	86.44	94.26
Metal sheets	99.47	92.19	93.54	99.62	73.99	99.78
Bare Soil	73.50	85.66	79.93	74.80	73.90	85.09
Bitumen	76.37	91.11	80.09	80.85	79.07	82.93
Bricks	82.69	80.63	72.34	87.25	78.24	92.03
Shadows	98.93	54.64	98.08	96.59	85.58	97.36
OA	89.38	86.30	90.27	89.91	85.06	**94.48**
AA	86.74	80.33	86.86	87.27	81.55	**92.68**
Kappa	0.858	0.817	0.870	0.865	0.803	**0.926**

The best results are highlighted in bold.

demonstrating that the SLCRF can outperform the other methods. In the proposed SLCRF method, the SL, the relationship construction, and the semisupervised constraints that are defined in the CRF are integrated to construct a unified objective function. The relationships between the image pixels in HSIs can be better described using the constructed spectral–spatial graph. Moreover, the CRF framework can enhance the performance of SL. Therefore, the SLCRF obtains much better HSI classification results.

4) In Table XI, the effects of training samples are analyzed by comparing the OA performance of SLCRF with CAE-SC as the percentage of the training data set is changed: 5%, 10%, 15%, and 20%. The classification accuracy of the proposed SLCRF method increases as the training samples increase. SLCRF provides higher accuracy than CAE-SC using different numbers of training samples.

D. Effectiveness of the Relationships' Construction Term

To evaluate the effectiveness of the relationships' construction term in the proposed method, we compare the SLCRF with the first reconstruction term and the CRF term, called RE-CRF, which is defined with the following equations:

$$\min_{\theta,\mathbf{h}} \frac{1}{b\times b\times d}\sum_{x=0}^{b-1}\sum_{y=0}^{b-1}\sum_{z=0}^{d-1}\left(X_{x,y,z}-X_{x,y,z}^{(M)}\right)^2 + \frac{\alpha}{2}\|\mathbf{W}\|_F^2 + \lambda_2\sum_{\boldsymbol{x}_i^{(M_1)}\in v} -I(t)\log\left(p\left(\boldsymbol{y}_i|\boldsymbol{x}_i^{(M_1)}\right)\right) + \lambda_2\eta\sum_{\boldsymbol{x}_i^{(M_1)}\in v}\sum_{\boldsymbol{x}_j^{(M_1)}\in N(\boldsymbol{x}_i^{(M_1)})} -\mathbf{h}_2\mathbf{S}_{2i,j}. \tag{24}$$

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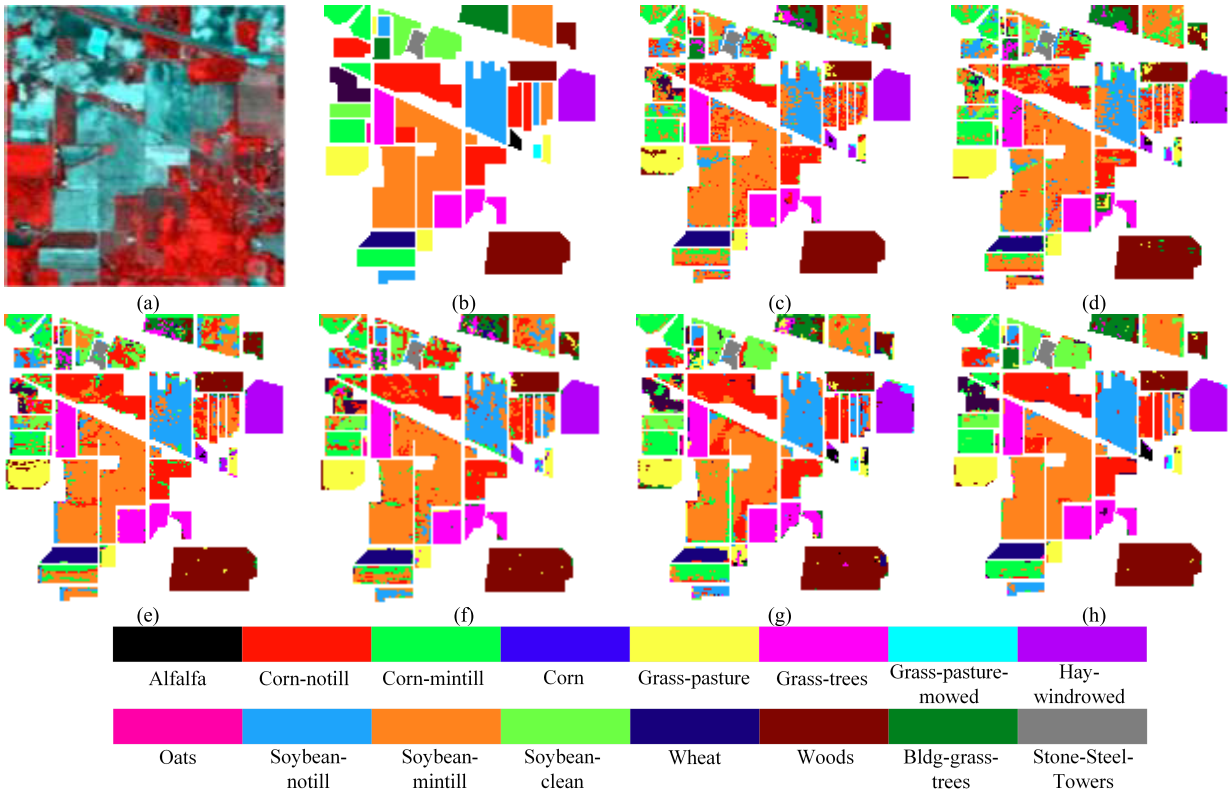


Fig. 2. Classification maps of the different methods with 5% labeled samples for the Indian Pines data set. (a) False color. (b) Ground truth. (c) PCA-SC. (d) LPP-SC. (e) CAE-SC. (f) SCAE-SC. (g) 3DCAE-CRF. (h) SLCRF.

TABLE X
SEMISUPERVISED CLASSIFICATION RESULTS (%) WITH 1% LABELED SAMPLES FOR THE HOUSTON DATA SET

Class	PCA-SC	LPP-SC	CAE-SC	SCAE-SC	3DCAE-CRF	SLCRF
Healthy Grass	83.70	70.62	91.21	94.61	83.99	89.14
Stressed Grass	92.66	77.14	93.80	93.22	89.08	94.46
Artificial Turf	99.56	96.60	98.52	99.41	67.21	96.61
Evergreen Trees	94.96	64.12	94.93	93.62	82.75	95.35
Deciduous Trees	56.01	41.22	60.58	64.88	65.40	73.22
Bare Earth	54.11	70.60	83.58	85.86	55.21	92.03
Water	90.11	66.54	89.35	91.63	30.46	87.10
Residential Buildings	71.04	71.72	76.98	74.44	61.08	82.12
Non-residential Buildings	90.45	89.45	93.30	92.92	86.23	94.78
Roads	45.34	49.32	48.24	44.63	58.58	63.66
Sidewalks	44.00	22.10	51.22	47.13	53.62	59.46
Crosswalks	0.00	0.13	0.33	0.00	0.00	15.02
Major Thoroughfares	50.51	61.97	65.43	73.72	56.49	77.69
Highways	45.36	74.29	69.34	78.61	51.28	88.29
Railways	79.67	85.51	93.84	96.51	74.79	98.10
Paved Parking Lots	72.02	75.18	77.44	84.13	59.69	86.12
Unpaved Parking Lots	0.00	29.93	31.29	2.72	2.83	75.56
Cars	36.96	46.45	38.30	43.96	62.49	64.53
Trains	69.63	61.74	74.64	77.44	61.20	87.53
Stadium Seats	85.05	88.82	87.70	89.42	70.96	93.17
OA	74.77	73.03	80.05	80.38	74.56	**85.64**
AA	63.06	62.17	71.00	71.44	58.67	**80.70**
Kappa	0.667	0.645	0.739	0.744	0.667	**0.813**

The best results are highlighted in bold.

From Table XII, the performance of the SLCRF significantly outperforms the method without the relationships' construction term in the three data sets, which further demonstrates that the relationships' construction term can provide a better solution to enhance the relationships of pixels.

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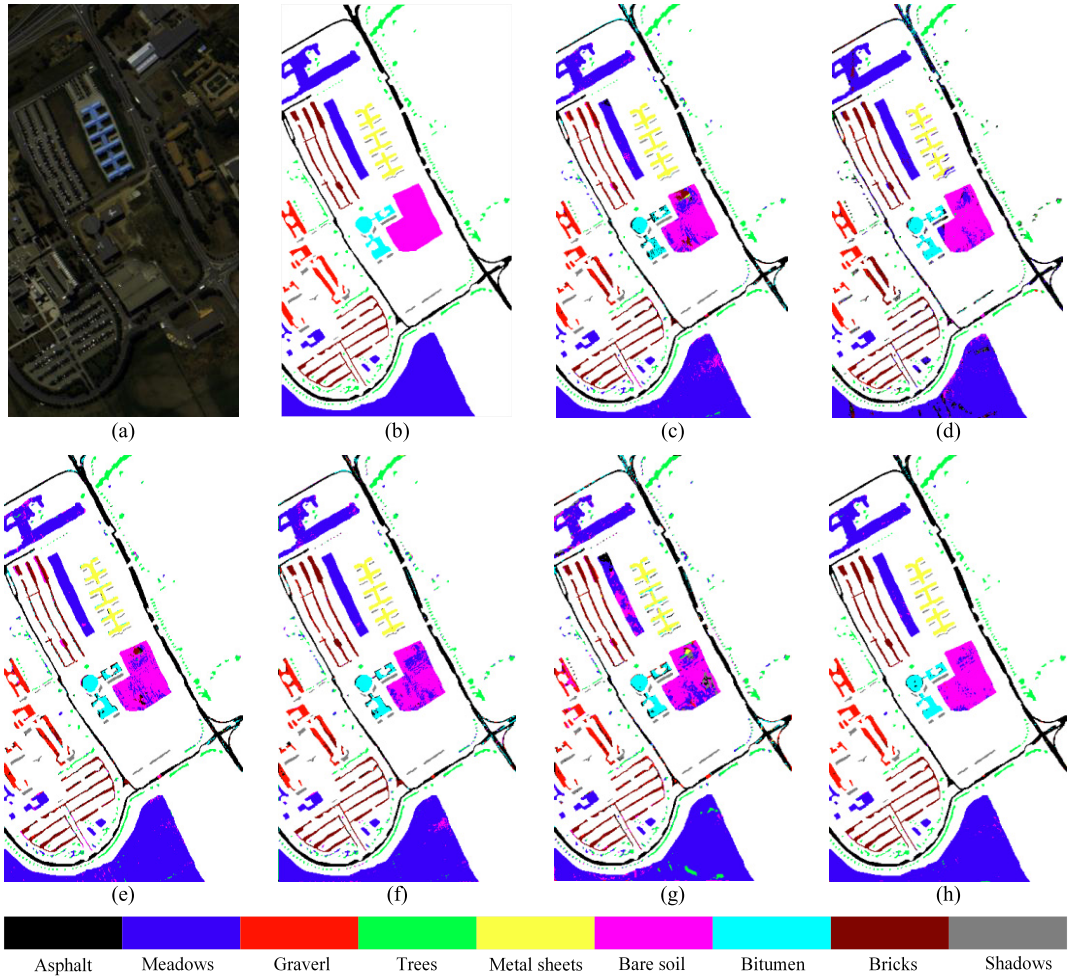


Fig. 3. Classification maps of the different methods with 5% labeled samples for the PaviaU data set. (a) False color. (b) Ground truth. (c) PCA-SC. (d) LPP-SC. (e) CAE-SC. (f) SCAE-SC. (g) 3DCAE-CRF. (h) SLCRF.

TABLE XI
PERFORMANCE COMPARISON WITH DIFFERENT PERCENTAGES' TRAINING SAMPLES WITH LABELED SAMPLES

Dataset	Method	5%	10%	15%	20%
Indian pines	CAE-SC	72.62	72.66	73.49	75.93
	SLCRF	**83.24**	**90.70**	**91.22**	**92.84**
PaviaU	CAE-SC	90.27	91.17	89.88	90.68
	SLCRF	**94.48**	**96.26**	**97.18**	**97.59**
Houston	CAE-SC	80.05	80.78	81.65	83.55
	SLCRF	**85.64**	**87.61**	**87.91**	**88.43**

TABLE XII
PERFORMANCE COMPARISON OF THE SL WITH LABELED SAMPLES

Dataset	RE-CRF	SLCRF
Indian pines	86.02	**90.70**
PaviaU	89.92	**94.48**
Houston	82.72	**85.64**

The best results are highlighted in bold.

E. Effectiveness of the CRF Term

To evaluate the effectiveness of the CRF in the proposed method, we compare the SLCRF with SL whose model does not consider the CRF framework and is defined with the following equations:

$$\min_{\theta,\mathbf{Z}} = \frac{1}{b\times b\times d}\sum_{x=0}^{b-1}\sum_{y=0}^{b-1}\sum_{z=0}^{d-1}\left(X_{x,y,z}-X_{x,y,z}^{(M)}\right)^2 + \frac{\alpha}{2}\|\mathbf{W}\|_F^2+\lambda_1\left\|\mathbf{X}^{(M_1)}-\mathbf{X}^{(M_1)}\mathbf{Z}\right\|_F^2+\lambda_1\beta\|\mathbf{Z}\|_1. \quad (25)$$

The features extracted by the SL are classified by using the random forest classifier (SL-RF). From the HSI classification results in Table XIII, we can see that the SLCRF significantly outperforms the model without the CRF in the three data sets. These results can further validate that CRF can provide a better solution to enhance SL's performance and incorporate the features for visual recognition.

F. Parameters Analysis

In the proposed method SLCRF, five parameters, including β, γ, η, λ_1, and λ_2, need to be tuned in each data set.

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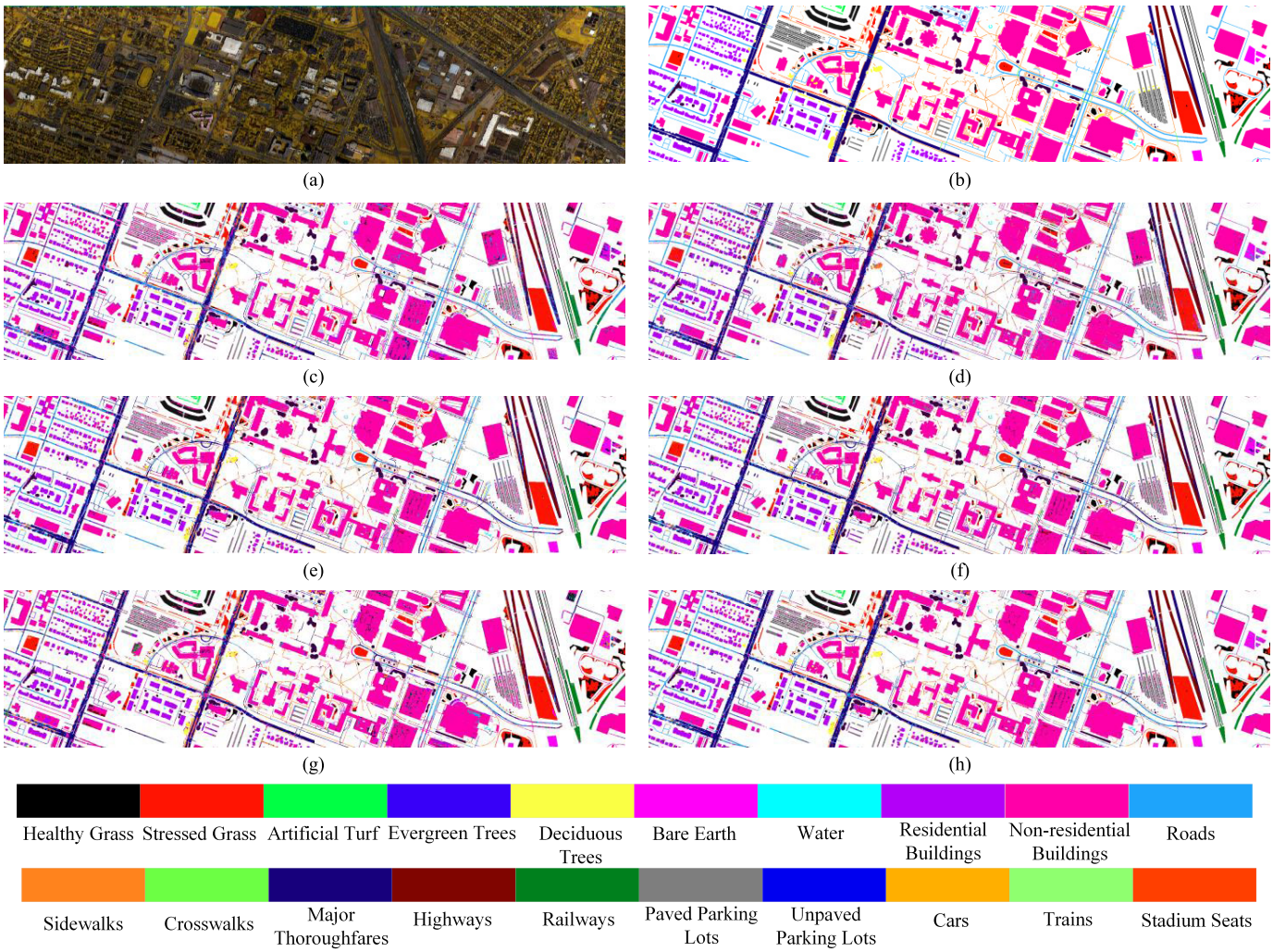


Fig. 4. Classification maps of the different methods with 5% labeled samples for the Houston data set. (a) False color. (b) Ground truth. (c) PCA-SC. (d) LPP-SC. (e) CAE-SC. (f) SCAE-SC. (g) 3DCAE-CRF. (h) SLCRF.

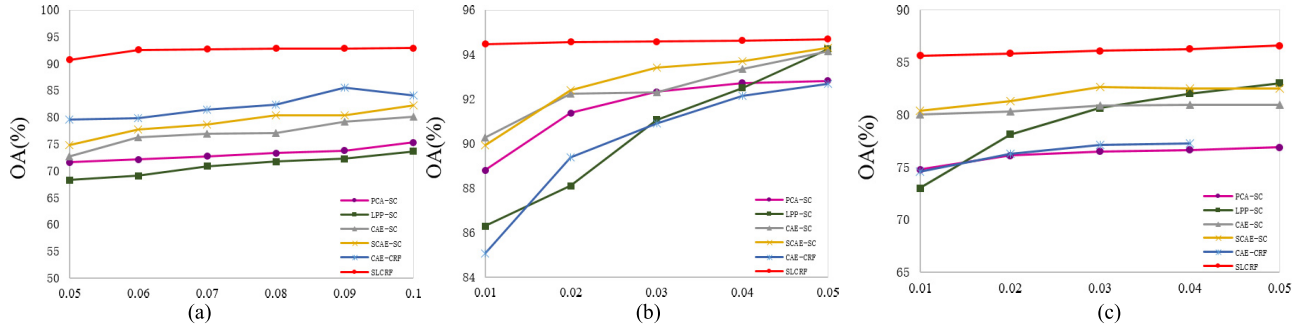


Fig. 5. Classification results of the PCA-SC, LPP-SC, CAE-SC, SCAE-SC, 3DCAE-CRF, and SLCRF with different percentages of labeled image samples. (a) Indian Pines data set. (b) PaviaU data set. (c) Houston data set.

The influences of these parameters on the HSI classification results are discussed.

The parameters λ_1 and λ_2 correspond to the pixels' relationships construction and CRF regularization, respectively. We have tuned these parameters, as shown in Fig. 6. Since every data set has its own distinctive data structure, the suitable parameters are different for each data set. From Fig. 6, it is also seen that the values of parameters γ and η have greater impacts on the image classification results than parameters β. The most suitable values of these parameters can be taken from the image classification performance. In conclusion, the best image classification results for the Indian Pines data set are achieved when the five parameters β, γ, η, λ_1, and λ_2 are close to 10^2, 10, 10^4, 10^3, and 10^{-3}, respectively. For the

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TABLE XIII
PERFORMANCE COMPARISON OF THE CRF WITH LABELED SAMPLES

Dataset	SL-RF	SLCRF
Indian pines	87.46	**90.70**
PaviaU	90.67	**94.48**
Houston	83.24	**85.64**

The best results are highlighted in bold.

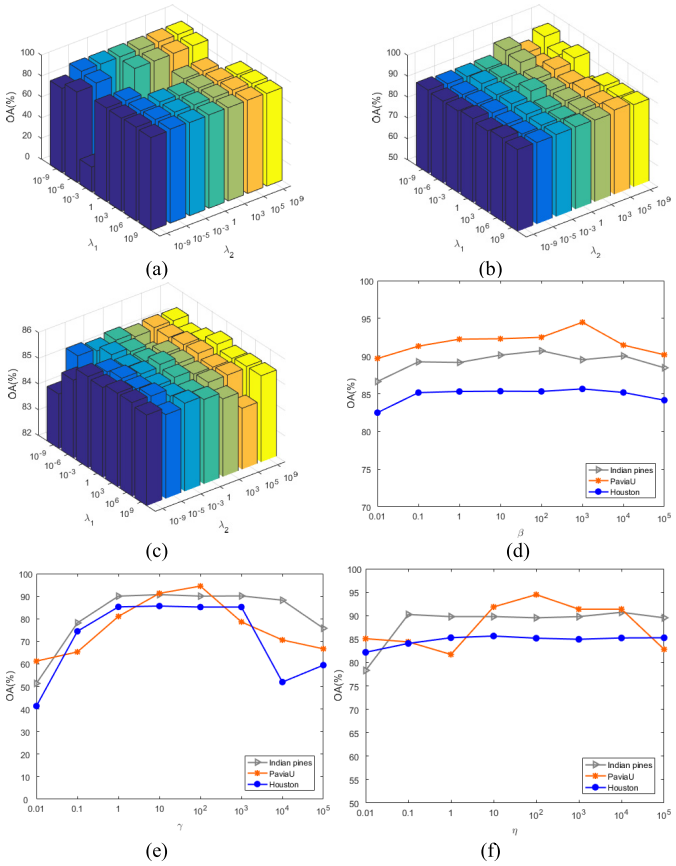


Fig. 6. Influences of the different parameter values on the HSI classification results. (a)–(c) Influences of λ_1 and λ_2 on the classification results for the Indian Pines, PaviaU, and Houston data sets. (d)–(f) Influences of different values of β, γ, and η on the classification results for the Indian Pines, PaviaU, and Houston data sets.

PaviaU data set, the best six parameters are those that are close to 10^3, 10^2, 10^2, 10^5, 10^3, and 10^{-9}, respectively. For the Houston data set, the best β, γ, η, λ_1, and λ_2 are to be 10^3, 10, 10, 10^3, and 10^{-9}, respectively.

V. CONCLUSION

In this article, a novel SL approach termed SLCRF is developed for the HSI classification. The main contributions of the SLCRF are its accurate integration of the probabilities of pixels belonging to different categories and the relationships between the neighboring pixels into the terms of the CRF. In this way, SLCRF can adopt semisupervised learning to alleviate the need for labeled samples. The proposed SLCRF method integrates the SL, relationship construction, and semi-supervised constraints that are defined in the CRF to build our objective function. The SLCRF uses an iterative optimization algorithm. Our proposed method is evaluated on three publicly used HSI classification data sets, and the experimental results show that the SLCRF method can outperform the related semisupervised HSI classification approaches.

In future work, SL will be integrated into other deep learning frameworks, such as graph convolutional network, to automatically obtain more discriminative and representative features of the HSI and further boost the HSI classification performance.

APPENDIX A
OPERATIONS FOR CAE

3-D Convolutional Layer: The 3-D convolution kernel in the mth convolution layer has the size of $P^{(m)} \times Q^{(m)} \times R^{(m)}$, where $R^{(m)}$ is the size of kernel along the spectral dimension. The value at position (x, y, z) on the jth feature map in the mth layer $(1 \le m < M_1 - l)$ $X^{(m)}_{x,y,z}$ can be described as

$$X^{(m,j)}_{x,y,z} = \psi\Big(b^{(m,j)} + \sum_{k=0}^{K^{(m)}-1}\sum_{p=0}^{P^{(m)}-1}\sum_{q=0}^{Q^{(m)}-1}\sum_{r=0}^{R^{(m)}-1} W^{(m,j,k)}_{p,q,r} X^{(m-1,k)}_{(x+p),(y+q),(z+r)}\Big) \quad (26)$$

where ψ is the activation function, j is the number of kernels in this layer, and $K^{(m)}$ is the number of feature cubes input to the mth layer. $W^{(m,j,k)}_{p,q,r}$ is the (p, q, r)th value of the kernel that is connected to the kth feature map in the previous layer, and $b^{(m,j)}$ is the bias.

3-D Max Pooling: The size of the feature map extracted by the convolutional layer is reduced using the form of pooling. Max pooling is the procedure that outputs the max value within the kernel size. For a $s \times s \times v$ window size neighbor denoted as Ω, the output of max pooling can be described as follows:

$$X^{(m)}_{x,y,z} = \max_{(x,y,z)\in\Omega} X^{(m-1)}_{x,y,z} \quad (27)$$

where $X^{(m-1)}_{x,y,z}$ $(1 \le m < M_1 - l)$ represents the features extracted from the 3-D convolutional layer and $X^{(m)}_{x,y,z}$ represents the features at position (x, y, z) after 3-D max pooling.

3-D Deconvolution Layer: The downsampled feature maps are upsampled through the deconvolution layer to map into a high-dimensional space. Compared with simple upsampling, the convolution is a trainable upsampling convolutional layer, which can maintain the important details. Let $(X^{(m-1)})'$ be the input of mth layer $(M_1 + l < m \le M)$ that have been padded with zeros in all the three dimensions. Then, the output of 3-D deconvolution layer is obtained by employing the convolutional kernels on $(X^{(m-1)})'$

$$X^{(m,j)}_{x,y,z} = \psi\Big(b^{(m,j)} + \sum_{k=0}^{K^{(m)}-1}\sum_{p=0}^{P^{(m)}-1}\sum_{q=0}^{Q^{(m)}-1}\sum_{r=0}^{R^{(m)}-1} W^{(m,j,k)}_{p,q,r}\Big(X^{(m-1,k)}_{(x+p),(y+q),(z+r)}\Big)'\Big). \quad (28)$$

Fully Connected Layer: At the last 3-D convolution layer, the feature map of the previous layer is flattened and fed into fully connected layers. There are $2l$ fully connected layers in

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CAE, where the encoder and the decoder each have l ($1< l < M_1$) layers

$$X_{x,y,z}^{(m)} = \sum_{i=1}^{L^{(m)}} \psi\left(b_i^{(m)} + W_i^{(m)} X_{x,y,z}^{(m-1)}\right) \tag{29}$$

where $L^{(m)}$ is the number of the hidden nodes at the mth layer ($M_1 - l \le m \le M_1 + l$). $W_i^{(m)}$ is the weights connecting the ith hidden node and the input node, and $b_i^{(m)}$ is the bias of the ith hidden node.

Appendix B
Optimization for SLCRF

Optimization for θ: θ is solved when $\mathbf{Z}$, $\mathbf{M}$, and $\mathbf{h}$ are fixed. The optimization problem defined in (18) is written as (19)

$$\min_{\theta} L(\theta) = \min_{\theta} \frac{1}{b \times b \times d} \sum_{x=0}^{b-1} \sum_{y=0}^{b-1} \sum_{z=0}^{d-1} \left(X_{x,y,z} - X_{x,y,z}^{(M)}\right)^2 + \frac{\alpha}{2}\|\mathbf{W}\|_F^2 + \lambda_1 \left\|\mathbf{X}^{(M_1)} - \mathbf{X}^{(M_1)}\mathbf{Z}\right\|_F^2 + \lambda_1 \beta \|\mathbf{Z}\|_1. \tag{30}$$

The optimization problem in (19) is solved using the stochastic gradient descent and backpropagation algorithm iteratively. The partial derivative for parameters of the mth ($1 \le m \le M$) layer of (19) is

$$\frac{\partial L(\theta)}{\partial \mathbf{W}^{(m)}} = \mathbf{U}^{(m)} (\mathbf{X}^{(m-1)})^T + \alpha \sum_{k=0}^{K^{(m)}-1} \sum_{p=0}^{P^{(m)}-1} \sum_{q=0}^{Q^{(m)}-1} \sum_{r=0}^{R^{(m)}-1} W_{p,q,r}^{(m,j,k)} \tag{31}$$

$$\frac{\partial L(\theta)}{\partial \mathbf{b}^{(m)}} = \mathbf{U}^{(m)} \tag{32}$$

where $\mathbf{W}^{(m)}$ and $\mathbf{b}^{(m)}$ represent the weights and bias in the mth layer, respectively, and $\mathbf{U}^{(m)}$ can be described as follows:

$$\begin{aligned}
\mathbf{U}^{(M)} &= -\frac{2}{b \times b \times d} \sum_{x=0}^{b-1} \sum_{y=0}^{b-1} \sum_{z=0}^{d-1} \left(X_{x,y,z} - X_{x,y,z}^{(M)}\right) \odot \psi'(\mathbf{a}^{(M)}) \\
\mathbf{U}^{(M_1)} &= ((\mathbf{W}^{(M_1+1)})^T \mathbf{U}^{(M_1+1)} \\
&\quad + \lambda_1 [2\mathbf{X}^{(M_1)}(\mathbf{I} - \mathbf{Z} - \mathbf{Z}^T - \mathbf{Z}\mathbf{Z}^T)]) \odot \psi'(\mathbf{a}^{(M_1)}) \\
&\quad + \lambda_2 \sum_{\boldsymbol{x}_i^{(M_1)} \in v} -\frac{1}{p\left(\boldsymbol{y}_i | \boldsymbol{x}_i^{(M_1)}\right)} \odot \Phi'(\mathbf{W}^{\mathbf{h}_1} \boldsymbol{x}_i^{(M_1)} + \mathbf{b}^{\mathbf{h}_1}) \\
&\quad \mathbf{U}^{(m)} (m \in [1, M_1 - l] \cup [M_1 + l, M]) \\
&= R_o(\mathbf{W}^{(m+1)} \otimes' \mathbf{U}^{(m+1)}) \odot \psi'(\mathbf{a}^{(m)}) \\
&\quad \mathbf{U}^{(m)} (m \in [M_1 - l, M_1 + l]) \\
&= ((\mathbf{W}^{(m+1)})^T \mathbf{U}^{(m+1)}) \odot \psi'(\mathbf{a}^{(m)})
\end{aligned} \tag{33}$$

where $\mathbf{a}^{(m)}$ represents the features that are not activated by ψ, R_o is the rotation of the input matrix by 180°, ψ' is the derivative of ψ, Φ' is the derivative of Φ, $\odot$ is the elementwise multiplication, $\otimes'$ is the transposed convolution operation, and $\mathbf{I}$ is the identity matrix.

Then, we would perform the update on each iteration

$$\mathbf{W}^{(m)} = \mathbf{W}^{(m)} - \delta_1 \frac{\partial L(\theta)}{\partial \mathbf{W}^{(m)}}, \mathbf{b}^{(m)} = \mathbf{b}^{(m)} - \delta_1 \frac{\partial L(\theta)}{\partial \mathbf{b}^{(m)}} \tag{34}$$

where δ_1 is the learning rate of the 3DCAE.

Optimization for Z: $\mathbf{Z}$ is computed when θ, $\mathbf{M}$, and $\mathbf{h}$ are fixed. As for $\mathbf{Z}$, the optimization problem defined in (18) is written as (20)

$$\begin{aligned}
&\min_{\mathbf{Z}} L(\mathbf{Z}) \\
&= \min_{\mathbf{Z}} \lambda_1 \left\|\mathbf{X}^{(M_1)} - \mathbf{X}^{(M_1)}\mathbf{Z}\right\|_F^2 \\
&\quad + \lambda_2 \eta \sum_{\boldsymbol{x}_i^{(M_1)} \in v} \sum_{\boldsymbol{x}_j^{(M_1)} \in N(\boldsymbol{x}_i^{(M_1)})} -\mathbf{h}_2 \mathbf{S}_{i,j} + \frac{\epsilon}{2} \left\| \mathbf{Z} - \mathbf{M} + \frac{\mathbf{T}}{\epsilon} \right\|_F^2.
\end{aligned} \tag{35}$$

According to the optimization condition, we have

$$\begin{aligned}
\frac{\partial L(\mathbf{Z})}{\partial \mathbf{Z}} &= 2\lambda_1 (\mathbf{X}^{(M_1)})^T \mathbf{X}^{(M_1)} (\mathbf{Z} - \mathbf{I}) \\
&\quad + \lambda_2 \eta \left(\frac{\operatorname{sgn}(\mathbf{Z} + \mathbf{Z}^T)}{2} \mathbf{h}_2{}^T + \mathbf{h}_2 \frac{\operatorname{sgn}(\mathbf{Z} + \mathbf{Z}^T)}{2} \right) \\
&\quad + \epsilon \left(\mathbf{Z} - \mathbf{M} + \frac{\mathbf{T}}{\epsilon} \right).
\end{aligned} \tag{36}$$

Then, we can obtain

$$\mathbf{Z} = \mathbf{Z} - \delta_2 \frac{\partial L(\mathbf{Z})}{\partial \mathbf{Z}} \tag{37}$$

where δ_2 is the learning rate of $\mathbf{Z}$.

Optimization for M, T: $\mathbf{M}$ can be solved when θ, $\mathbf{Z}$, and $\mathbf{h}$ are fixed. For the $\mathbf{M}$ subproblem, the optimization problem defined in (18) is written as (21):

$$\min_{\mathbf{M}} L(\mathbf{M}) = \min_{\mathbf{M}} \beta \|\mathbf{M}\|_1 + \tfrac{\epsilon}{2} \left\|\mathbf{Z} - \mathbf{M} + \tfrac{\mathbf{T}}{\epsilon}\right\|_F^2. \tag{38}$$

Note that (21) is the l_1–l_2 term, and it can be directly solved using the shrinkage operator [34]. This is as follows:

$$\mathbf{M}^k(\cdot, i) = \max\left\{0, \left\|\mathbf{Q}^{(k)}(\cdot, i)\right\|_2 - \frac{\beta}{\epsilon}\right\} \frac{\mathbf{Q}^{(k)}(\cdot, i)}{\left\|\mathbf{Q}^{(k)}(\cdot, i)\right\|_2} \tag{39}$$

where

$$\mathbf{Q}^{(k)} = \mathbf{Z}^{(k)} + \frac{\mathbf{T}^{(k)}}{\epsilon}. \tag{40}$$

Meanwhile, $\mathbf{T}$ can be updated by

$$\mathbf{T}_{\text{new}} = \mathbf{T}_{\text{old}} + \epsilon(\mathbf{Z} - \mathbf{M}). \tag{41}$$

Optimization for h: For the $\mathbf{h}$ optimization, with the original objective function that is defined in (23), we can get the following equations:

$$\begin{aligned}
\frac{\partial L(\mathbf{h})}{\partial \mathbf{W}^{\mathbf{h}_1}} &= \sum_{\boldsymbol{x}_i^{(M_1)} \in v} -\frac{1}{p\left(\boldsymbol{y}_i | \boldsymbol{x}_i^{(M_1)}\right)} \odot \Phi'(\mathbf{W}^{\mathbf{h}_1} \boldsymbol{x}_i^{(M_1)} + \mathbf{b}^{\mathbf{h}_1}) \left(\boldsymbol{x}_i^{M_1}\right)^T \\
\frac{\partial L(\mathbf{h})}{\partial \mathbf{b}^{\mathbf{h}_1}} &= \sum_{\boldsymbol{x}_i^{(M_1)} \in v} -\frac{1}{p\left(\boldsymbol{y}_i | \boldsymbol{x}_i^{(M_1)}\right)} \odot \Phi'\left(\mathbf{W}^{\mathbf{h}_1} \boldsymbol{x}_i^{(M_1)} + \mathbf{b}^{\mathbf{h}_1}\right)
\end{aligned} \tag{42}$$

and

$$\frac{\partial L(\mathbf{h})}{\partial \mathbf{h}_2} = \sum_{\boldsymbol{x}_i^{(M_1)} \in v} \sum_{\boldsymbol{x}_j^{(M_1)} \in N(\boldsymbol{x}_i^{(M_1)})} -\mathbf{S}_{i,j}. \tag{43}$$

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Therefore, we can update $\mathbf{h}_1$ and $\mathbf{h}_2$ as follows:

$$\mathbf{h}_1 = \mathbf{h}_1 - \tau \frac{\partial L(\mathbf{h})}{\partial \mathbf{h}_1}, \quad \mathbf{h}_2 = \mathbf{h}_2 - \tau \frac{\partial L(\mathbf{h})}{\partial \mathbf{h}_2} \tag{44}$$

where τ is the step size.

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